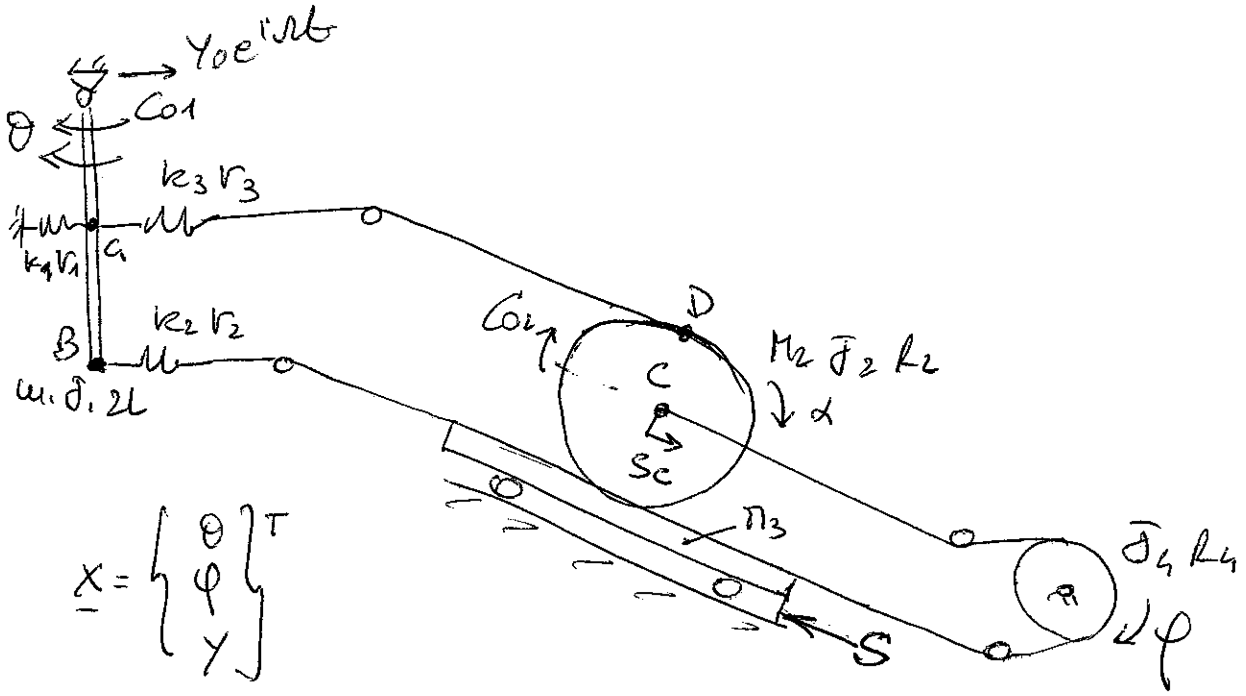
2 dic 2010 - B



$$\underline{X} = \begin{Bmatrix} \theta \\ \varphi \\ Y \end{Bmatrix}^T$$

$$E_C = \frac{1}{2} M_1 \dot{x}_G^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} M_2 V_C^2 + \frac{1}{2} J_2 \dot{\alpha}^2 + \frac{1}{2} M_3 V^2 + \frac{1}{2} J_4 \dot{\varphi}^2$$

$x_G = -L\theta + Y$

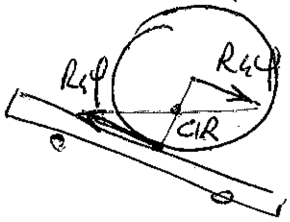
$S = R_4 \varphi$

$S_C = R_4 \varphi$

$\alpha = \dfrac{2R_4\varphi}{R_2}$

$\underline{Y}_{MF} = \{ x_G \ \theta \ S_C \ \alpha \ S \ \varphi \}^T$

$[M_F] = diag(M_1 \ J_1 \ M_2 \ J_2 \ M_3 \ J_4)$



$2R_4\varphi$ = spost. relativo centro disco - carrello

$$[\Lambda_M] = \begin{bmatrix} -L & 0 & 1 \\ 1 & 0 & 0 \\ 0 & R_4 & 0 \\ 0 & 2R_4/R_2 & 0 \\ 0 & R_4 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{cases} \Delta l_1 = Y - L\theta \\ \Delta l_2 = -S - x_B \\ \Delta l_3 = -x_G + x_D \\ \quad x_D = S_C + R_2\alpha \end{cases}$$

$$U_{EL} = \frac{1}{2} k_1 \Delta l_1^2 + \frac{1}{2} k_2 \Delta l_2^2 + \frac{1}{2} k_3 \Delta l_3^2$$

$$\begin{cases} \Delta l_1 = Y - L\theta \\ \Delta l_2 = -Y + 2L\theta - R_4\varphi \\ \Delta l_3 = -Y + L\theta + 3R_4\varphi \end{cases}$$

$$[\Lambda_{EL}] = \begin{bmatrix} -L & 0 & 1 \\ 2L & -R_4 & -1 \\ L & 3R_4 & -1 \end{bmatrix}$$

$$V_g = -m_1 g L \cos\theta \rightarrow \frac{\partial^2 V_g}{\partial \theta^2} = m_1 g L \cos\theta \qquad [K_g] = \begin{bmatrix} M_1 g L & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\delta L = C_{01}\,\delta\theta + C_{02}\,\delta\alpha + R_x\,\delta y =$$

$$= C_{01}\,\delta\theta + C_{02}\,\frac{2R_4}{R_2}\,\delta\varphi + R_x\,\delta y$$

$$Q_{C1} = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} C_{01} e^{i\Omega t}$$

$$Q_{C2} = \begin{Bmatrix} 0 \\ \frac{2R_4}{R_2} \\ 0 \end{Bmatrix} C_{02} e^{i\Omega t}$$

$$\underline{Q}_v = \{ R_x \}$$